Math 401 Review 1 Solutions.

1a) $\int_0^2 (x-2)^2\,dx$

$\int_0^2 (x^2-4x+2)\,dx$

$\frac{x^2}{2} - 2x^2 + 2x \Big|_0^2$

$2 - 8 + 4 - 0$

$= 0$

b) $\int \frac{2x^2 - 7x + 3}{3x^2}\,dx$

$\int \frac{2x^2}{3x^2} - \frac{7x}{3x^2} + \frac{3}{3x^2}\,dx$

$\int \frac{2}{3} - \frac{7}{3}\cdot\frac{1}{x} + \frac{1}{x^2}\,dx$

$= \frac{2}{3}x - \frac{7}{3}\ln|x| - \frac{1}{x} + C$

c) $\int \frac{4\sec^2 x}{\tan x}\,dx$

Let $u = \tan x$, $du = \sec^2 x\,dx$ $\Rightarrow \int \frac{4}{u}\,du$

$= 4\ln|u| + C$

$= 4\ln|\tan x| + C$

d) $\int \frac{\cos x \cdot e^{\sin x}}{\sqrt{e^{\sin x}}}\,dx$

Let $u = e^{\sin x}$

$du = e^{\sin x}\cdot\cos x\,dx$

$\Rightarrow \int \frac{1}{\sqrt{u}}\,du = \int u^{-1/2}\,du$

$= \frac{2}{1}u^{1/2} = 2\sqrt{e^{\sin x}} + C$

e) $\int x(x^2+4)^2\,dx$

expand or u-sub

Expand: $\int x(x^4 + 8x^2 + 16)\,dx$

$\int x^5 + 8x^3 + 16x\,dx$

$= \frac{x^6}{6} + 2x^4 + 8x^2 + C$

u-sub: Let $u = x^2+4$

$du = 2x\,dx \Rightarrow \frac{1}{2}\,du$

$\Rightarrow \int \frac{1}{2}\cdot u^2\,du$

$\Rightarrow \frac{1}{2}\cdot\frac{u^3}{3} + C$

$\Rightarrow \frac{1}{6}(x^2+4)^3 + C$

1 f) $\int \frac{1+x^5}{x^4}\,dx$

$\Rightarrow \int \frac{1}{x^4} + \frac{x^5}{x^4}\,dx$

$\Rightarrow \int (x^{-4} + x)\,dx$

$\Rightarrow \frac{x^{-3}}{-3} + \frac{x^2}{2} + C$

$\Rightarrow -\frac{1}{3x^3} + \frac{x^2}{2} + C$

g) $\int \frac{x^4}{1+x^5}\,dx$

Let $u = 1+x^5$

$du = 5x^4\,dx \Rightarrow \frac{1}{5}du = x^4\,dx$

$\Rightarrow \int \frac{1}{5} \cdot \frac{1}{u}\,du$

$= \frac{1}{5}\ln|u| + C$

$= \frac{1}{5}\ln|1+x^5| + C$

h) $\int \frac{x^2}{\sqrt{1+x^3}}\,dx$

Let $u = 1+x^3$

$du = 3x^2\,dx$

$\frac{1}{3}du = x^2\,dx$

$\Rightarrow \int \frac{1}{3} \cdot \frac{1}{\sqrt{u}}\,du$

$\Rightarrow \frac{1}{3}\int u^{-1/2}\,du = \frac{1}{3} \cdot \frac{2}{1}u^{1/2} + C$

$= \frac{2}{3}\sqrt{1+x^3} + C$

i) $\int \frac{1}{(1+x^2)\tan^{-1}x}\,dx$

Let $u = \tan^{-1}x$

$du = \frac{1}{1+x^2}\,dx$

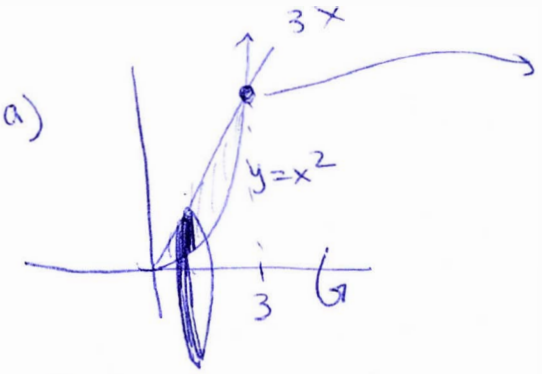
$\Rightarrow \int \frac{1}{u}\,du$

$= \ln|u| + C$

$\Rightarrow \ln|\tan^{-1}x| + C$

2)

a)



$x^2 = 3x$

$x^2 - 3x = 0$

$x(x-3) = 0$

$x = 0$ or $x = 3$

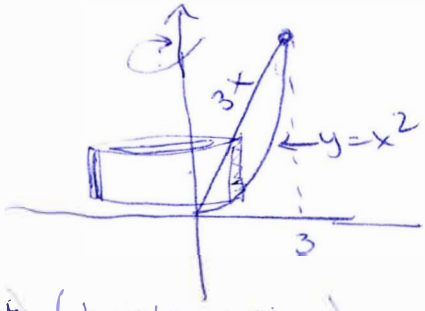
Washer method

$$V = \int_a^b \pi (R^2 - r^2)\,dx$$

$$= \pi \int_0^3 (3x)^2 - (x^2)^2\,dx$$

$$= \pi \int_0^3 (9x^2 - x^4)\,dx$$

b) About y-axis



Shell Method

$$V = 2\pi \int_a^b (\text{radius of shell})(\text{height of shell})$$

$$= 2\pi \int_0^3 x\,(3x - x^2)\,dx$$

$$= 2\pi \int_0^3 (3x^2 - x^3)\,dx$$

c) About $y = -4$

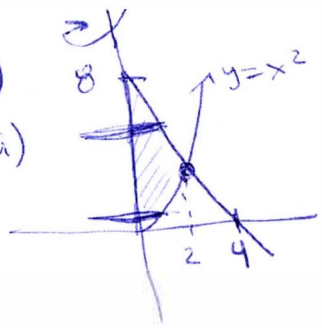
~~Shell~~ Washer (Same picture as "a")

$$V = \int_0^3 \pi \left[(3x+4)^2 - (x^2+4)^2\right] dx$$

d) About $x = 3$; Shell:

$$V = 2\pi \int_0^3 \underbrace{(3-x)}_{\text{radius}} \underbrace{(3x - x^2)}_{\text{height}}\,dx$$

3)



a) $V = \pi \int_a^b (R^2 - r^2)\,dy$

$V = \pi \int_0^2 (\sqrt{y})^2\,dy + \pi \int_2^8 \left(4 - \frac{y}{2}\right)^2 dy$

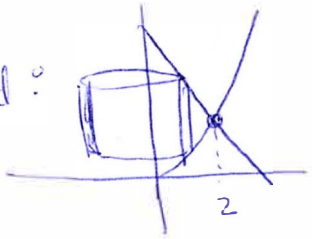
$V = \pi \int_0^2 y\,dy + \pi \int_2^8 \left(4 - \frac{y}{2}\right)^2 dy$

$y = 8 - 2x$

$2x = 8 - y$

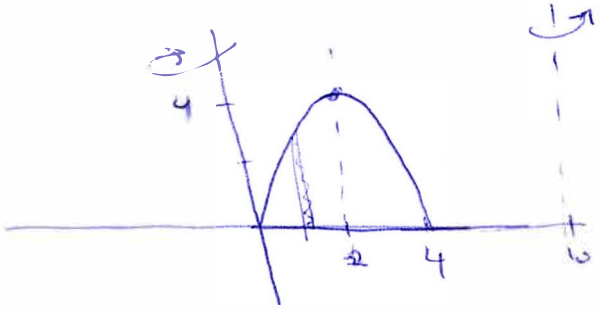
$x = 4 - \frac{y}{2}$

b) Shell Method:



$V = 2\pi \int_0^2 x\left(-2x + 8 - x^2\right)dx$

4)



a) y-axis : use shells

$V = 2\pi \int_0^4 x\left(4x - x^2\right)dx$

b) $x = 10$ ← vertical line

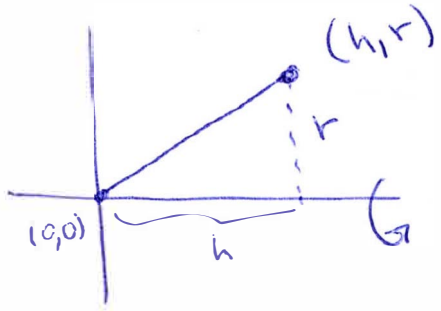
Radius $= 10 - x$

$V = 2\pi \int_0^4 (10 - x)\left(4x - x^2\right)dx$

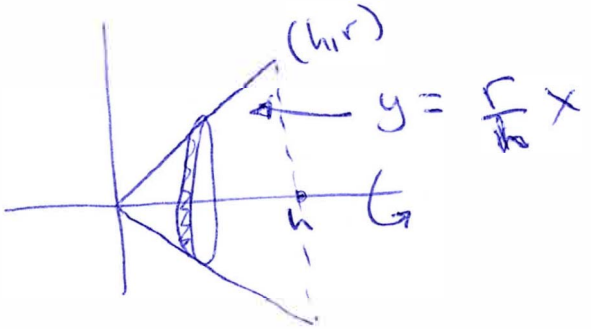
c) $x = 2$

$2\pi \int_0^2 (2 - x)\left(4x - x^2\right)dx$

5)



$\Rightarrow$

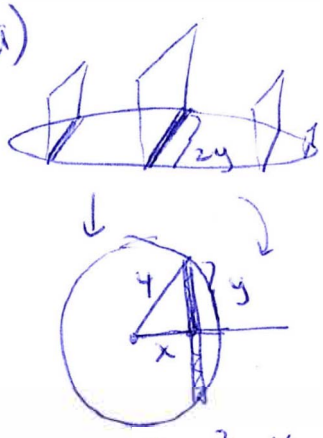


$$V = \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx$$

$$= \pi \int_0^h \frac{r^2}{h^2} x^2 \, dx$$

$$= \pi \frac{r^2}{h^2} \cdot \frac{x^3}{3}\Big|_0^h = \frac{1}{3}\pi r^2 h$$

6) a)



$x^2 + y^2 = 16$

$y = \sqrt{16 - x^2}$

Side $= 2y = 2\sqrt{16 - x^2}$

Area of $\square$ = (Side)^2 $= (2\sqrt{16 - x^2})^2$

Done in class

$$V = \int_a^b (\text{Area of square})\, dx$$

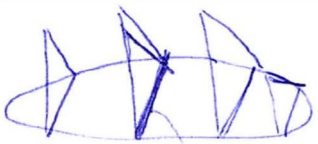
$$V = \int_{-4}^{4} 4(16 - x^2)\, dx = \int_{-4}^{4} 64 - 4x^2 \, dx$$

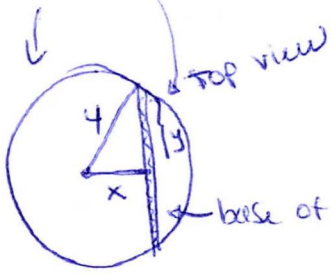
$$= 64x - \frac{4}{3}x^3 \Big|_{-4}^{4}$$

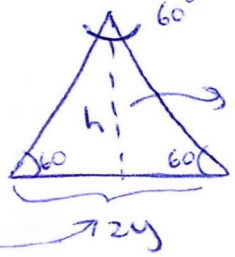
$$256 - \frac{256}{3} - \left(-256 + \frac{256}{3}\right)$$

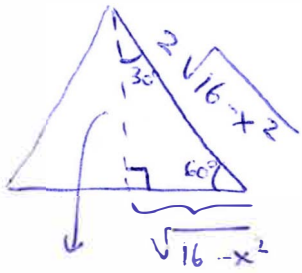
$$512 - \frac{512}{3} = \frac{1024}{3}$$

6 b) Equilateral triangles









$x^2+y^2 = 16$

$y = \sqrt{16-x^2}$

base $= 2y = 2\sqrt{16-x^2}$

$h = (\text{short side}) * \sqrt{3}$

$h = \sqrt{3} \cdot \sqrt{16-x^2}$

$$\text{Vol} = \int_{-4}^{4} (\text{Area of triangle})\, dx$$

$$= \int_{-4}^{4} \frac{1}{2}\left(2\sqrt{16-x^2}\right)\cdot\left(\sqrt{3}\cdot\sqrt{16-x^2}\right) dx$$

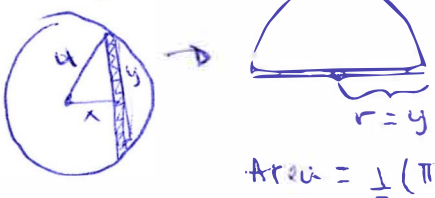
$$= \sqrt{3}\int_{-4}^{4} (16-x^2)\, dx = \sqrt{3}\left[16x - \frac{x^3}{3}\right]_{-4}^{4}$$

$$= \sqrt{3}\left[64 - \frac{64}{3} - \left(-64 + \frac{64}{3}\right)\right]$$

$$= \sqrt{3}\left[128 - \frac{128}{3}\right] = \frac{256\cdot\sqrt{3}}{3}$$

6 c) Semicircles





$y = \sqrt{16-x^2}$

$\text{Area} = \frac{1}{2}(\pi r^2)$

$= \frac{1}{2}\pi y^2$

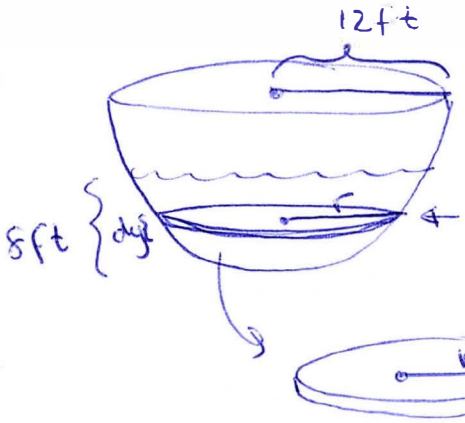
$$\text{Vol} = \int_{-4}^{4} (\text{Area of Semicircle})\, dx$$

$$\int_{-4}^{4} \frac{1}{2}\pi\left(\sqrt{16-x^2}\right)^2 dx$$

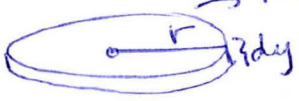
$$\vdots$$

$$\frac{128\pi}{3}$$

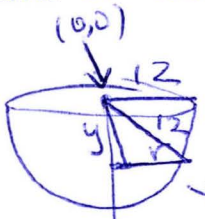
7)



← cross section is a disk: Volume of Disk = $\pi r^2 h$ w.r.t. dy



→ find r in terms of y:



$\rightarrow y^2 + r^2 = 12^2$

$\leftarrow r = \sqrt{144 - y^2}$

∴ Vol:

$$V = \int_4^{12} \pi (r)^2 dy$$

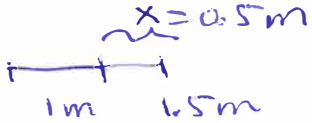
→

$$= \int_4^{12} \pi \left(\sqrt{144 - y^2}\right)^2 dy$$

Since (0,0) is at the top

⋮

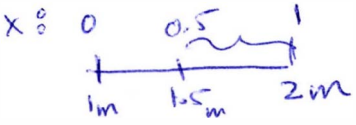
8) $F = kx$



$40 = k(0.5)$

$80 = k$

∴ $F = 80x$



$\rightarrow W = \int_{1/2}^{1} 80x\,dx$

$= 40x^2 \Big|_{1/2}^{1} = 40 - 10$

$= 30$ Joules.

9)

(0,0)

50m

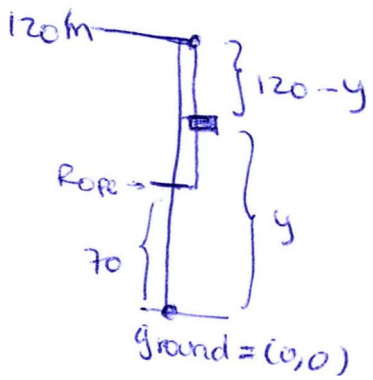
Ground = 120m

$$\int_0^{50} 0.5y\,dy = \int_0^{50} \frac{1}{2}y\,dy$$

$$= \frac{1}{4}y^2\Big|_0^{50}$$

$$= \frac{2500}{4} = 625$$

OR with (0,0) at the bottom:



$$W = \int_{70}^{120} (0.5\ \text{lb/m})(120 - y)\,dy \quad \text{meters} \qquad 0.5 = \frac{1}{2}$$

$$W = \int_{70}^{120} \left(60 - \frac{1}{2}y\right) dy$$

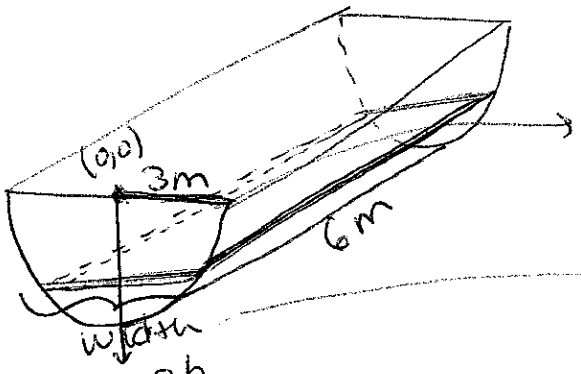
$$= 60y - \frac{1}{4}y^2\Big|_{70}^{120}$$

$$= 7200 - 3600 - (4200 - 1225)$$

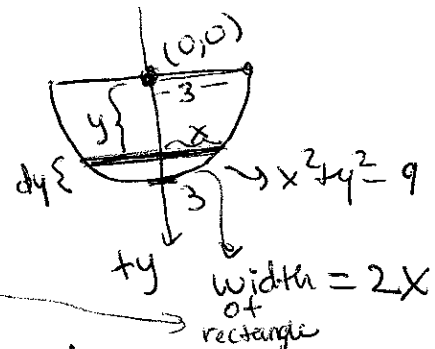
$$3600 - 2975$$

$$= 625$$

10)



Front view:



$$W = \int_a^b \rho \cdot g \,(\text{area of cross section})\,(\text{distance out of the tank}) \cdot dx \text{ or } dy$$

$$W = \int_0^3 (1000)(9.8)\,(\text{length} \cdot \text{width}) \cdot y \cdot dy$$

$$\int_0^3 (9800)\left(6 \cdot 2\sqrt{9-y^2}\right) \cdot y\,dy$$

$$\int_0^3 9800 \cdot 12 \cdot y \cdot \sqrt{9-y^2}\,dy$$

$$117600 \int_0^3 y\sqrt{9-y^2}\,dy$$

Let $u = 9-y^2$

$du = -2y\,dy$

$-\frac{1}{2}du = y\,dy$

$$-58800 \int \sqrt{u}\,du \qquad y^{1/2}$$

$$-58800 \cdot \frac{2}{3} u^{3/2}$$

$$-39200\,(9-y^2)^{3/2}\Big|_0^3$$

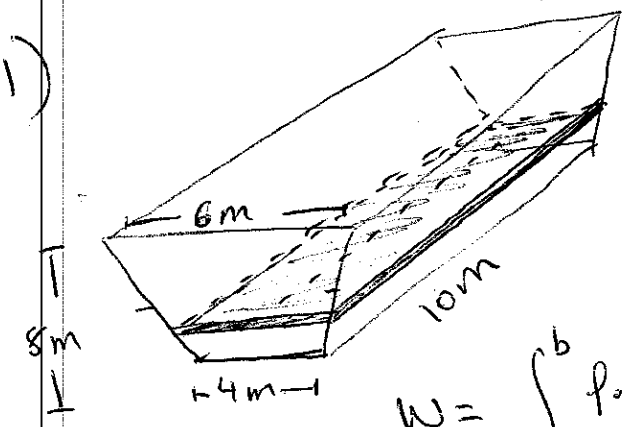
$9^{3/2} = (\sqrt{9})^3$

$$-39200\,(0) + 39200\,(9)^{3/2}$$

$$= 39200\,(27)$$

$$= 1{,}058{,}400 \text{ J}$$

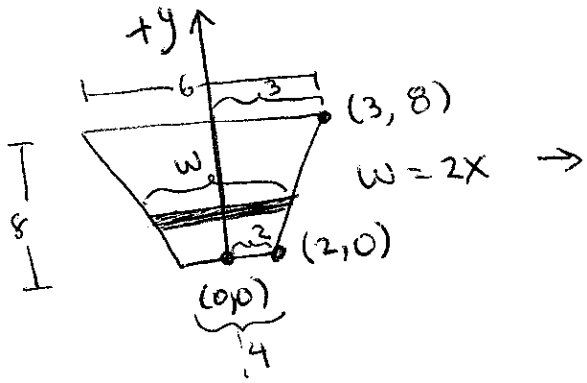
11)



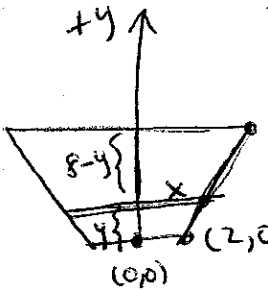
$$W = \int_a^b \rho \cdot g \cdot (\text{Area of Crossection})(\text{Distance out of trough})\, dy$$

rectangle → $A = l \cdot w$ (l = 10m)

with (0,0) bottom middle:



$w = 2x \rightarrow$



→ find line:

$m = \frac{8-0}{3-2} = \frac{8}{1}$

$y - y_1 = m(x - x_1)$

$y - 0 = 8(x-2)$

$y = 8x - 16$

$w = 2x$ → solve for x

$\frac{y+16}{8} = x$

a) $W = \int_0^8 (890)(9.8)(10)\left(2\left(\frac{y+16}{8}\right)\right)\cdot(8-y)\,dy$ (10 = length, width = $2\left(\frac{y+16}{8}\right)$)

$W = 21805 \int_0^8 (y+16)(8-y)\,dy = \ldots = 1.305 \times 10^7$

b) 2 meters from the top means there is no oil from 6m to 8m

$\therefore W = \int_0^6 (890)(9.8)(10)\left(2\left(\frac{y+16}{8}\right)\right)\cdot(8-y)\,dy$ ← The distance out of the tank is the same whether the tank is full or not.

$W = 21805 \int_0^6 (y+16)(8-y)\,dy$

$\vdots = 12{,}036{,}360$